

6(a)(i)	any four from: 1 water soluble ; 2 (so) easily transported around the cell ; 3 (ATP) loses, P_i / phosphate group ; 4 hydrolysed by ATPase / (ATP is) stable molecule ; 5 to release energy, immediately / in small packets or <i>ref. to</i> 30.5 kJ (mol⁻¹) ; 6 can be, recycled / regenerated or $ATP \rightleftharpoons ADP + P_i$;	4
6(a)(ii)	any two from: 1 unwinding DNA ; 2 activating (RNA) nucleotides ; 3 <i>ref. to</i> mRNA synthesis ; 4 peptide bond formation / joining amino acids ; 5 AVP ; e.g. post translation modification / amino acid activation / moving ribosomes along mRNA / movement of mRNA from nucleus	2

Question	Answer	Marks								
6(b)(i)	<table><tr><th>statement</th><th>letter</th></tr><tr><td>the site of the Krebs Cycle</td><td>A</td></tr><tr><td>a phospholipid bilayer impermeable to H+ ions</td><td>E / D</td></tr><tr><td>the site of translation</td><td>C</td></tr></table> ;; <i>all correct = 2 marks</i> <i>1 or 2 correct = 1 mark</i>	statement	letter	the site of the Krebs Cycle	A	a phospholipid bilayer impermeable to H+ ions	E / D	the site of translation	C	2
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6b(ii)	$(2 \times \pi \times 0.36^2) + (2 \times \pi \times 0.36 \times 5.25)$ or $2\pi \times 0.36 (0.36 + 5.25) ;$ 12.7 or 12.68 or 12.69 ; <i>accept 3.14 or π in workings</i>	2								
6(b)(iii)	1 (more cristae results in) more, ETC / (named) carrier proteins / ATP synthase ; or (more cristae results in) more, oxidative phosphorylation / chemiosmosis / ATP synthesis ; 2 (because) cardiac muscle must undergo continuous contractions / AW ;	2								